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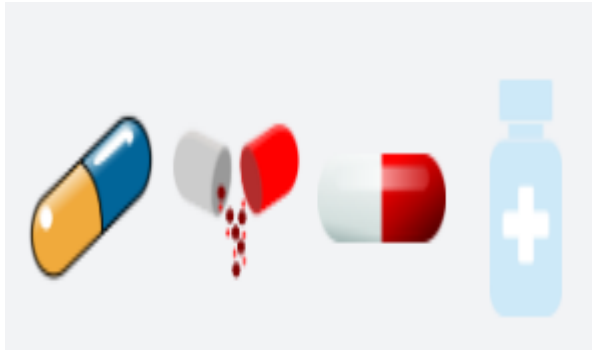

LEC-4

Cholinergic Antagonist Drugs

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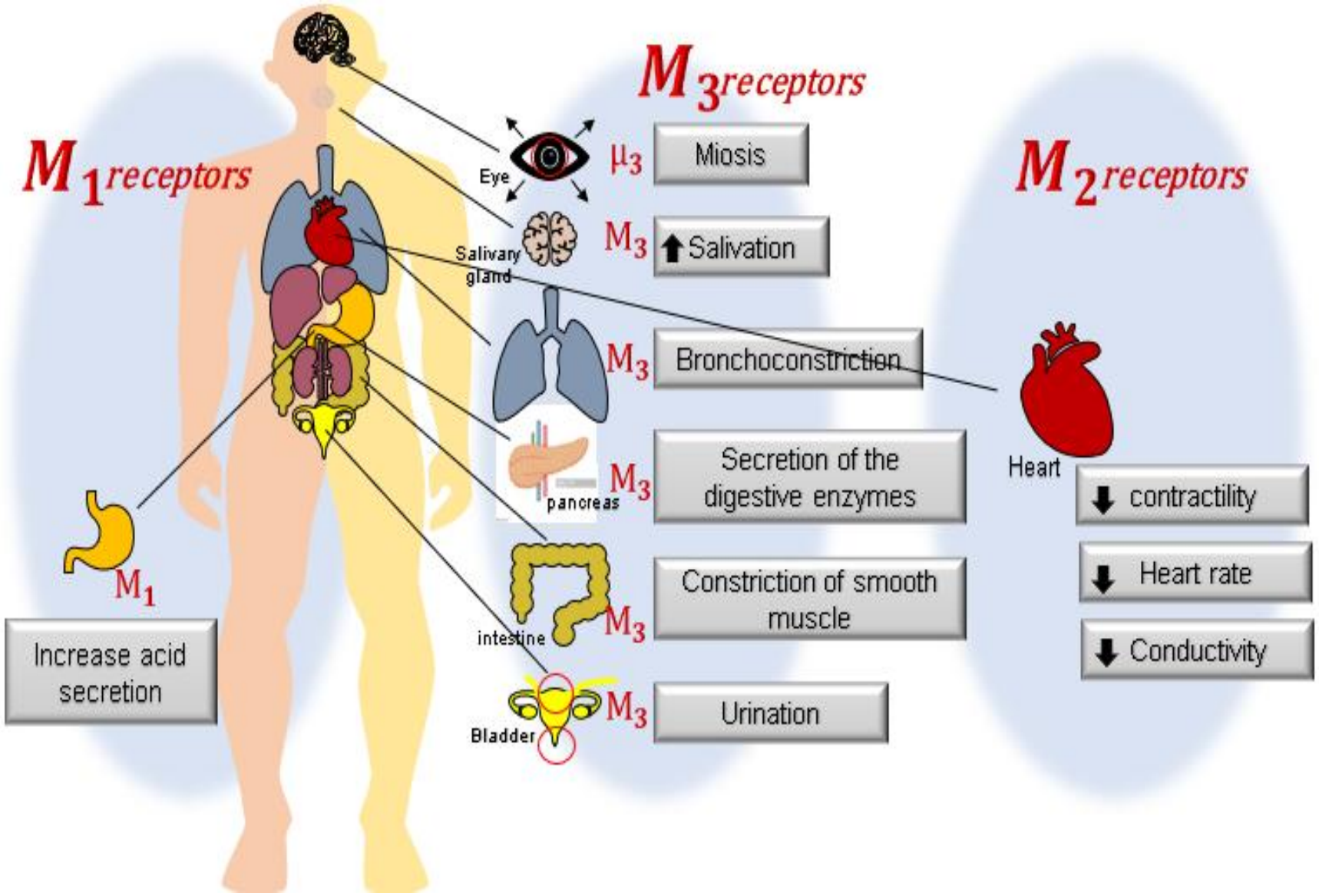
Learning Objectives

By the end of this lecture, you should be able to

- Describe the mechanism of action of each class at their respective receptor sites..
- Explain the effects of atropine and other antimuscarinic drugs..
- Identify Therapeutic Uses and Clinical Indications.
- Recognize Adverse Effects and Contraindications

Muscarinic Receptors

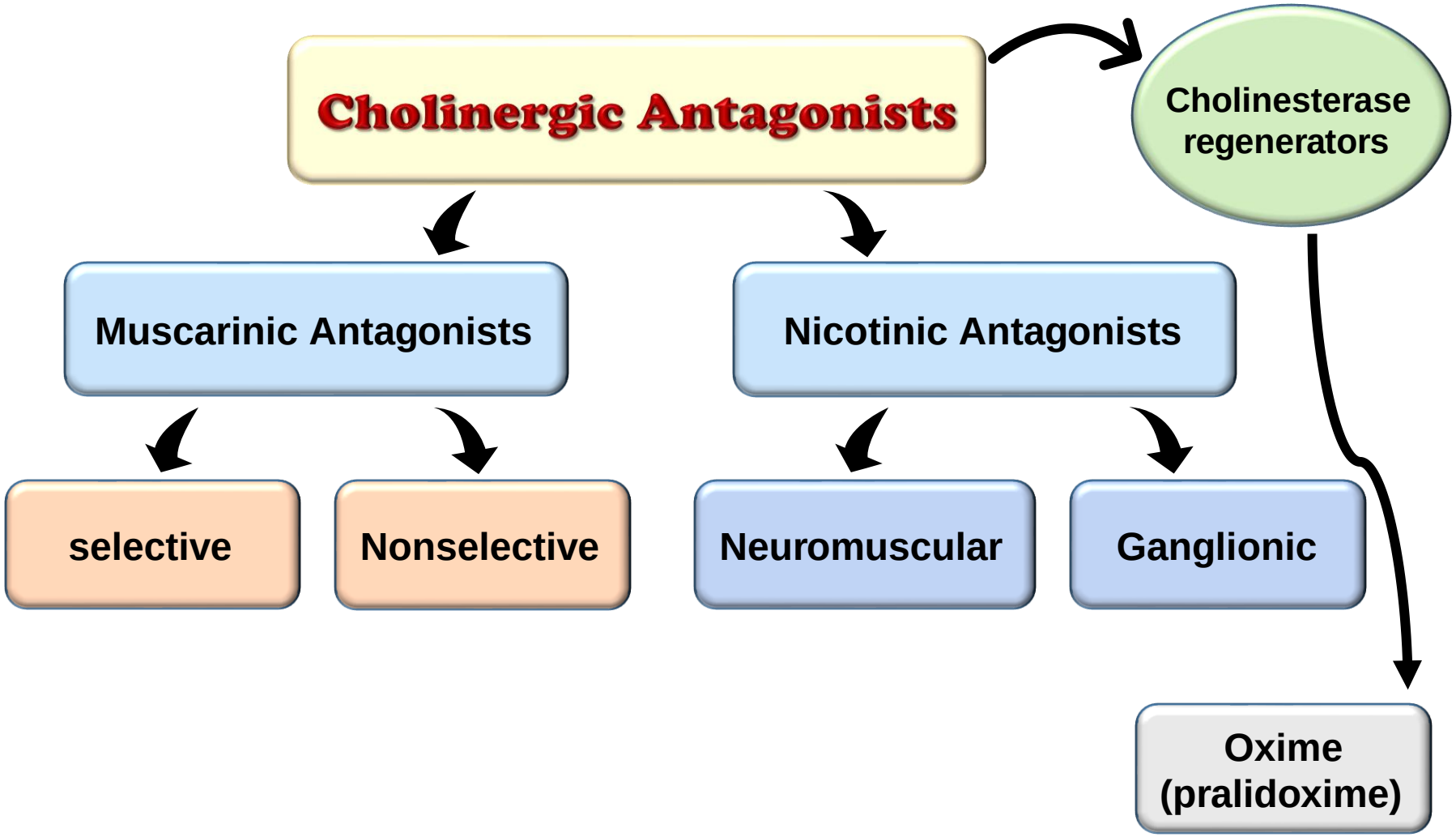


Cholinergic antagonists

Cholinergic antagonists are drugs that **bind to cholinoceptors (muscarinic or nicotinic)** and **inhibit the actions of acetylcholine (ACh)** and other cholinergic agonists.

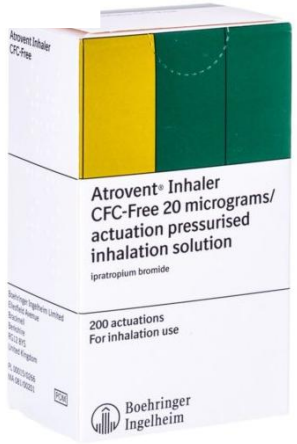
They are classified into three main categories:

1. **Antimuscarinic (anticholinergic or parasympatholytic) agents** ïselectively block **muscarinic receptors**.
2. **Ganglionic blockers** ïact on **nicotinic receptors** at autonomic ganglia.
3. **Neuromuscular blockers** ïinterfere with transmission at **skeletal neuromuscular junctions** (nicotinic receptors).


Cholinergic Antagonists
Cholinesterase regenerators
Muscarinic Antagonists
Nicotinic Antagonists
selective
Nonselective
Neuromuscular
Ganglionic
Oxime (pralidoxime)

Cholinergic Antagonists

- Muscarinic Antagonists
  - selective
    - M1 selective: Pirenzepine
    - M3 selective: Tolterodine
  - Nonselective
    - Tertiary amine: Atropine, Benzotropine, Scopolamine, Cyclopentolate, tropicamide
    - Quaternary amine: Ipratropium, Oxybutynin
- Nicotinic Antagonists



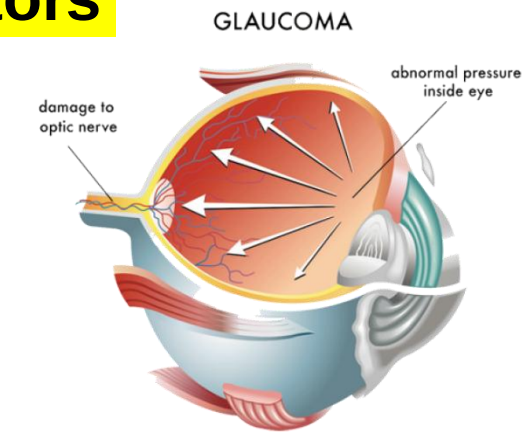
Effects of Muscarinic Antagonists

➢ Effects on the CNS.

- Muscarinic receptors (M_1, M_2, M_4, M_5) are widely distributed in the brain and spinal cord, where they regulate arousal, memory, motor control, and autonomic integration.
- Muscarinic antagonists that cross the blood–brain barrier (BBB) — mainly tertiary amines such as atropine and scopolamine — can therefore exert significant CNS effects, depending on dose and lipid solubility.
- By contrast, quaternary ammonium compounds (e.g., ipratropium, glycopyrrolate) are polar and do not penetrate the CNS, so they lack central effects.

Effects on the eye.

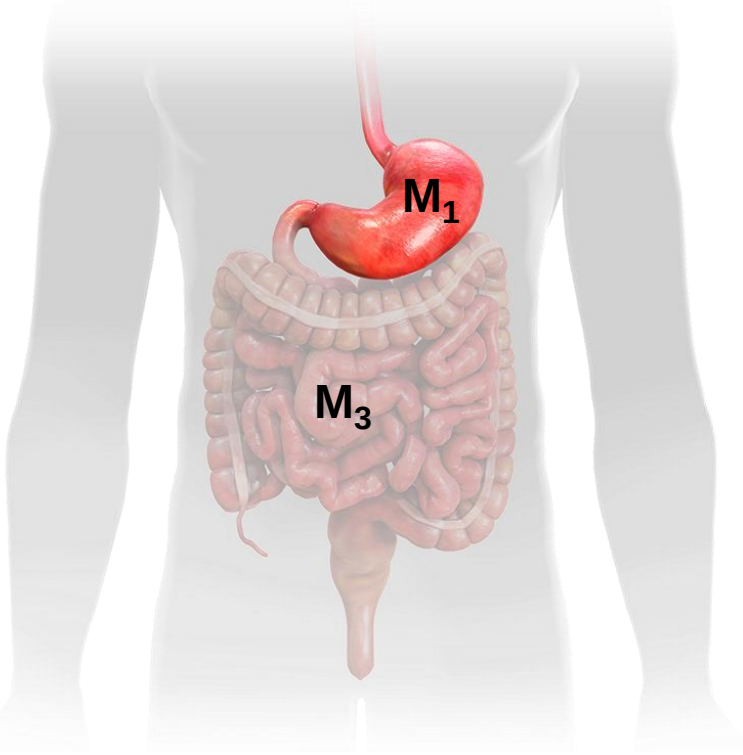
- The pupil is dilated (*mydriasis*) by atropine administration, and becomes unresponsive to light. Relaxation of the ciliary muscle causes paralysis of accommodation (*cycloplegia*) so that near vision is impaired.
- Intraocular pressure may rise; although this is unimportant in normal individuals, it can be dangerous in patients suffering from narrow-angle glaucoma.
- Eye effects is due to block of M_3 receptors



➢Effects on the gastrointestinal tract.

üGastrointestinal motility is inhibited by atropine, although this requires larger doses than the other effects listed, and is not complete.

- **Atropine** is used in pathological conditions in which there is increased gastrointestinal motility; agents selective for M_3 receptors, which are being developed, may be preferable .
  - **Pirenzepine**, owing to its selectivity for M_1 receptors, inhibits gastric acid secretion in doses that do not affect other systems



➢Inhibition of secretions.

- Salivary, lacrimal, bronchial and sweat glands are **inhibited** by very low doses of atropine.
- Gastric secretion is only slightly reduced.
- Mucociliary clearance in the bronchi is **inhibited**
- Block of M_1, M_3 receptors

➢Effects on smooth Muscle

- **Relax smooth** muscles in the respiratory and gastrointestinal tracts.
- They **delay gastric** emptying and decrease esophageal and gastric emptying
- These drugs also cause **bronchial dilation.**

➢ Anesthetic premedication :

- To dry secretions: for example **atropine, scopolamine.**
- To block the vagal slowing of the heart that can occur with general anesthesia

➢ Neurological :

- **Prevention of *motion sickness***: for example **scopolamine**
- ***Parkinsonism*** especially to counteract movement disorders caused by antipsychotic drugs: for example **benzhexol**, **benztropine**.

Effects of Antimuscarinic Drugs on Body Organs

Organ System	Normal Parasympathetic (Muscarinic) Effect	Effect of Antimuscarinic Drugs	Clinical Significance
CNS	Cholinergic transmission; arousal, memory, coordination	Sedation (scopolamine), excitation or delirium (atropine), improved motor control in Parkinsonism (benztropine)	Motion sickness, Parkinson's disease, anesthesia adjunct
Eye	Miosis (pupil constriction), accommodation for near vision	Mydriasis, ↑ intraocular pressure	Ophthalmic exams (tropicamide), contraindicated in glaucoma
Heart	↓ Heart rate (via M₂ receptors on SA node)	Tachycardia, ↑ conduction	Treat bradycardia (atropine)
Blood Vessels	Minimal direct parasympathetic effect	No direct effect; possible cutaneous vasodilation ("atropine flush") at high doses	Not used for vascular effects
Lungs	Bronchoconstriction, ↑ bronchial secretions	Bronchodilation, ↓ mucus secretion	COPD, asthma (ipratropium)
Gastrointestinal Tract	↑ Motility, ↑ secretions, relaxation of sphincters	↓ Motility, ↓ secretion, constipation, dry mouth	Antispasmodic (hyoscyamine)

Effects of Antimuscarinic Drugs on Body Organs

Organ System	Normal Parasympathetic (Muscarinic) Effect	Effect of Antimuscarinic Drugs	Clinical Significance
Salivary Glands	Salivation	Dry mouth (xerostomia)	Side effect; may cause speech/swallowing difficulty
Sweat Glands	Sweating, thermoregulation	↓ Sweating → hyperthermia, especially in hot environments	Toxic effect, esp. in children & elderly
Urinary Bladder	Contraction of detrusor, relaxation of sphincter → urination	Urinary retention, ↓ bladder contraction	Overactive bladder (oxybutynin)
Genital Tract	Erection (via parasympathetic tone)	↓ Erectile function	Usually mild; not therapeutic
Exocrine Glands (lacrimal, nasal)	Secretion (tears, mucus)	Dry eyes, dry nasal passages	Side effect, esp. with long-term use

Atropine

- Belladonna alkaloid
- Non-selective (blocks all muscarinic receptors)
- Readily absorbed from the GIT
- Partially metabolized and 60% excreted unchanged in urine.

- Cross BBB and cause CNS stimulation.
- The half-life is about 2 hours (but the effect of atropine on the eye may last from three days to one week).

Atropa Belladonna

Main Indications:

- As adjunct for anesthesia (reduce secretions, bronchodilatation)
- Anticholinesterase poisoning
- Treatment of Bradycardia (e.g. after MI)
- Gastrointestinal hypermotility (antispasmodic, antidiarrheal)

Main side effects:

Urinary retention, dry mouth, blurred vision, constipation, tachycardia, CNS disturbance (excitation, restlessness, agitation, hallucination).

Contraindications:

- Glaucoma (especially closed-angle)
- Benign prostatic hypertrophy (BPH)
- Used with caution infants with fever

A-Atropine

System	Effects
Eye	Mydriasis (pupil dilation), ↑ intraocular pressure (contraindicated in glaucoma)
GI tract	Decreases motility and spasm; ineffective for ulce
Cardiovascular	Low dose → bradycardia; high dose → tachycardia (dose-dependent)
Secretions	Inhibits salivary, sweat, and lacrimal glands → dry mouth, hot skin, risk of hyperthermia

Adverse Effects

Dry mouth, blurred vision, urinary retention, tachycardia, constipation, CNS disturbances (hallucinations, delirium). Toxicity may be reversed by physostigmine.

Therapeutic Uses

- Mydriatic and cycloplegic in ophthalmology
- Antispasmodic for GI disorders
- Treatment of bradycardia
- Preoperative antisecretory agent
- Antidote for organophosphate poisoning, cholinesterase inhibitor overdose, and muscarinic mushroom poisoning.

B. Scopolamine (Hyoscine)

- Similar to atropine but more CNS-active and longer-acting.
- Produces sedation, amnesia, and at higher doses, euphoria.

Uses

- Most effective agent for motion sickness (transdermal patch up to 3 days)
- Prevents postoperative nausea and vomiting

Side Effects

Blurred vision, confusion, urinary retention, dry mouth, mydriasis.

Cyclopentolate and Tropicamide

Tertiary amines developed for ophthalmic use (as mydriatic agents). Ophthalmic duration of action is much shorter than atropine and scopolamine (≈ 6 hours).

Ipratropium and Tiotropium

Quaternary ammonium compounds similar to atropine, but do not inhibit mucociliary clearance from bronchi.

Used by inhalation as bronchodilators for the treatment of asthma and COPD.

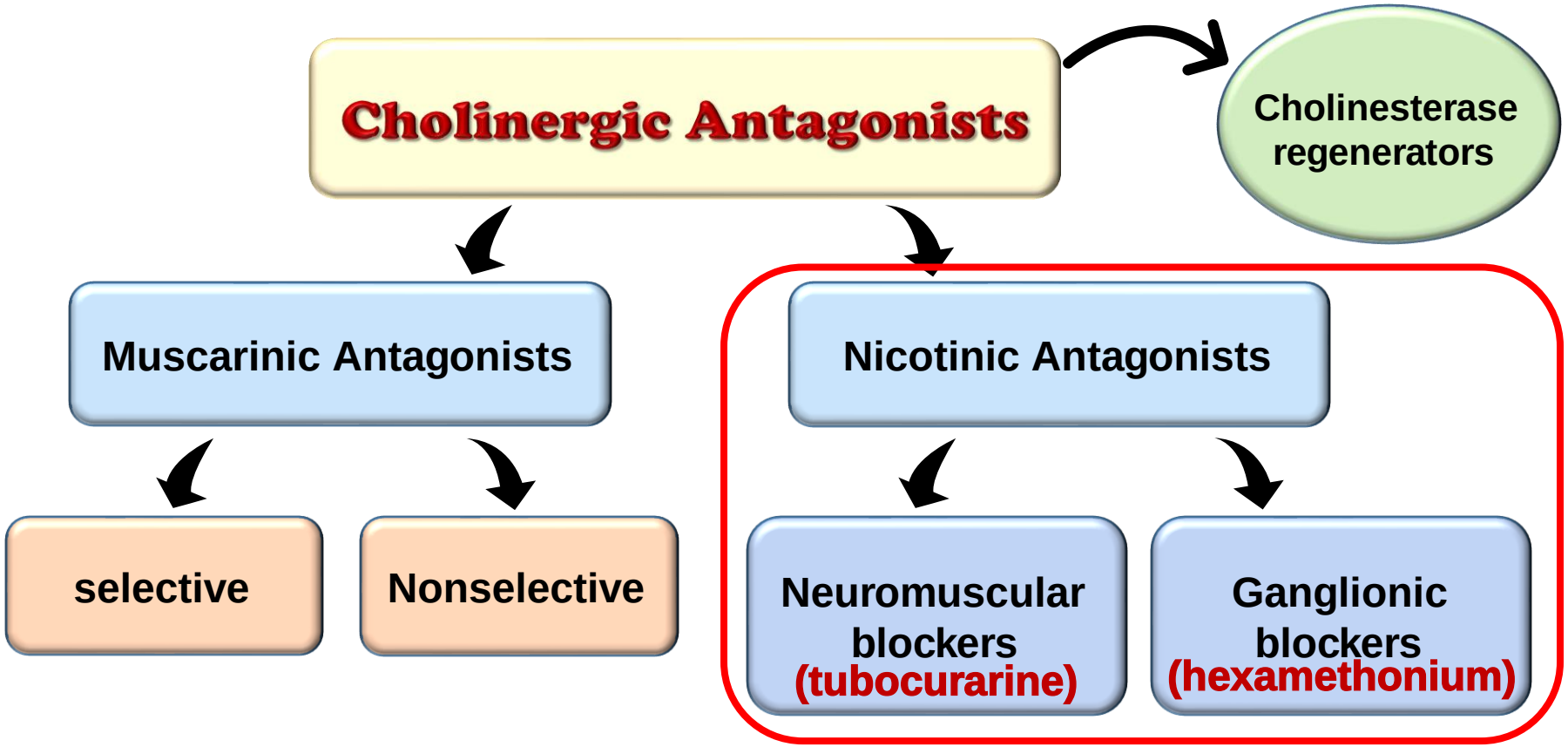
Tiotropium has longer duration of action and used once daily to control COPD

Selective muscarinic antagonists:

Currently used muscarinic antagonists show little subtype selectivity, but few drugs are selective muscarinic blockers, they include:

1- Pirenzepine (M_1-selective) Used to treat *peptic ulcer* disease by suppressing gastric acid secretion.

2- Oxybutynin (M_3-selective) used for treating *urinary incontinence*.


Cholinergic Antagonists
Cholinesterase regenerators
Muscarinic Antagonists
Nicotinic Antagonists
selective
Nonselective
Neuromuscular blockers (tubocurarine)
Ganglionic blockers (hexamethonium)

•Nicotinic Antagonists

- ➢Ganglion-blocking drugs
- ➢Neuromuscular-blocking drugs.

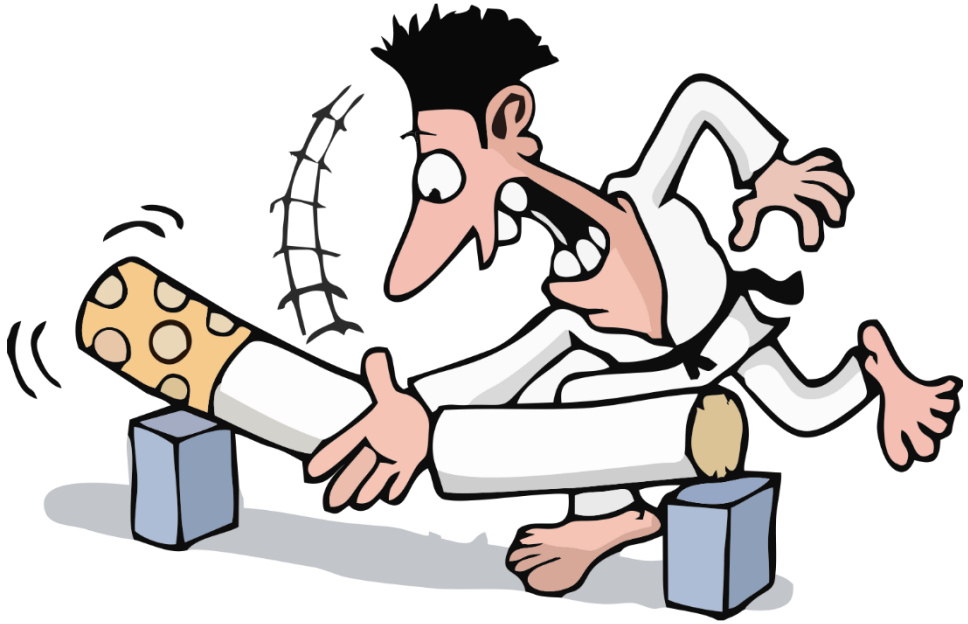
•Ganglion-Blocking Drugs

- Blockers of ganglionic nicotinic receptors act like competitive pharmacologic antagonists. These drugs were the first successful agents for the treatment of hypertension. Hexamethonium, mecamylamine, and several other ganglion blockers were extensively used for this disease.
- **Unfortunately, the adverse effects of ganglion blockade in hypertension are so severe (both sympathetic and parasympathetic divisions are blocked) that patients were unable to tolerate them for long periods**
- **Trimethaphan**

➢ **It is poorly lipid-soluble, inactive orally, and has a short half-life. It was used intravenously to treat severe accelerated hypertension (malignant hypertension)**

Varenicline

- (a partial agonist given by mouth), and **mecamylamine**, a nicotinic ganglion blocker that enters the CNS, have all been shown to have some benefit in smoking cessation.



❑Neuromuscular blockers (muscle relaxants)

➢ Depolarizing muscle relaxants

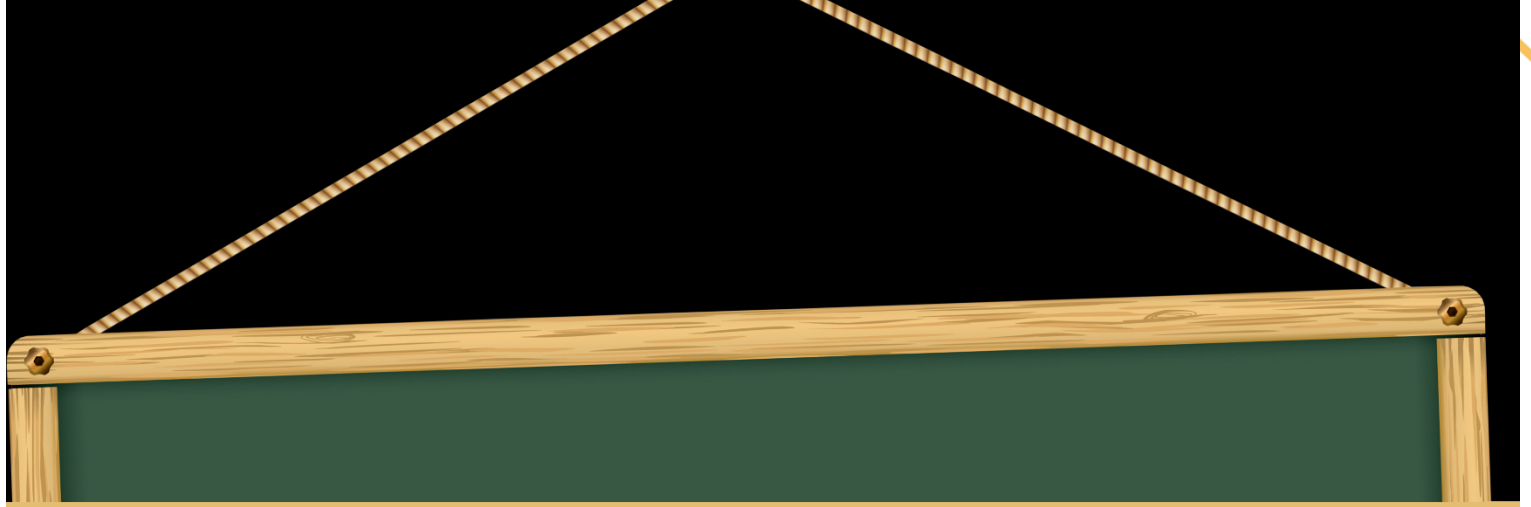
- **Succinylcholine**

➢Nondepolarizing muscle relaxants

- **Short acting Mivacurium**
- **Intermediate acting Atracurium**
- **Long acting Pancuronium**

❑Primary uses of neuromuscular-blocking drugs:

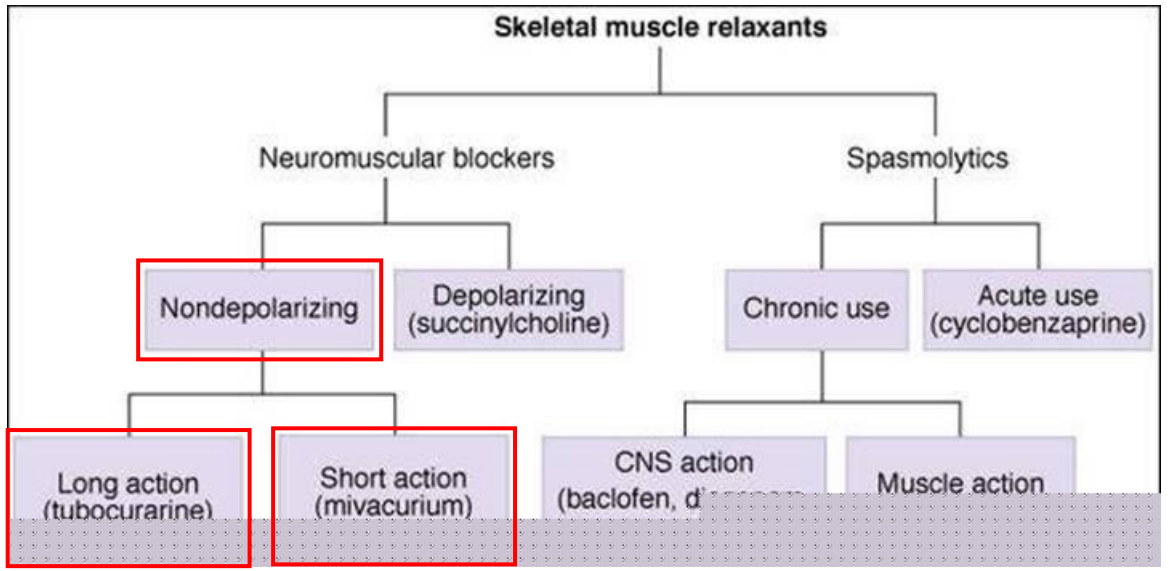
1. Skeletal muscle relaxation facilitating tracheal intubation
2. Skeletal muscle relaxation to improve intraoperative surgical conditions



Nondepolarizing muscle relaxants

What is the mechanism of action?

- Compete with Ach at the binding sites
- Do not depolarized the motor endplate
- Act as competitive antagonist
- Excessive concentration causing channel blockade
- Act at presynaptic sites, prevent movement of Ach to release sites

Tubocurarine

- is a prototype. It produces a competitive block at the end plate nicotinic receptor, causing flaccid paralysis that lasts 30 to 60 minutes (longer if large doses have been given). Pancuronium, atracurium, vecuronium, and several newer drugs are shorter-acting, nondepolarizing blockers.